

1 Motivation & Research questions

 Random dimension removal shows redundancy, but not how task-relevant information is distributed.

RQ1 Within a task, is information concentrated in a few dimensions or broadly distributed?

RQ2 Across tasks, are important dimensions task-specific or transferable?

RQ3 What explains differences across model families?

2 Evaluation Setup

11 embedding models

 **Retrieval-optimized**

- GTR-T5-Large
- BGE-M3
- GTE-Base
- MxBai-Embed-Large
- GTE-Large

 **Instruction-conditioned**

- RoBERTa-InBedder
- Qwen3-Embedding-0.6B
- Stella EN 400M
- Instructor-Large

 **General-purpose Backbone**

- RoBERTa-Large
- BART-Base

 34 MTEB tasks

7 categories: Classification, Clustering, Retrieval, STS, Pair Classification, Reranking, Summarization

 **Chunk-based dimension selection**

Retained dimensions: 16, 32, 64, 128, 256, 512

3 Diagnostic 1: Within-task dimension selection

Embedding vector (d dimensions)



Optimized selection

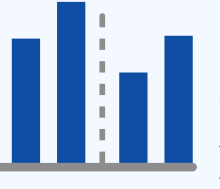
Select top-ranked chunks for the task.

Random selection

Randomly select chunks.

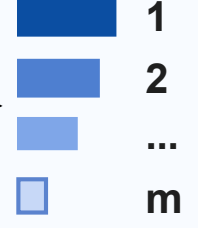
Anti-optimized selection

Select lowest-ranked chunks for the task.

**Retention relative to full dimensions**
(percentage of full-dimensional performance)

 Optimized-random gap diagnoses information concentration.
Smaller gap→broader distribution;
Larger gap→stronger concentration.

4 Diagnostic 2: Cross-task transfer



Rank chunks on source task, select top chunks

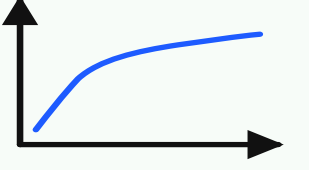
Evaluate on target task

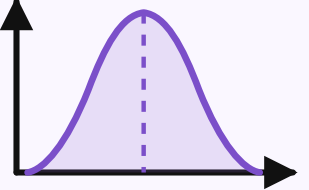
 Rank agreement across tasks

 Transfer retention

 Tests whether chunk-importance rankings are task-specific and whether tasks still share redundant structure.

5 Mechanism Analysis

Top-k concentration curves


Normalized Shannon Entropy


 Examines how evenly chunk importance is distributed and helps explain family differences.

6 Key Findings

- 1** Retrieval-optimized Embedders and most Instruction-conditioned Embedders: small optimized-random gaps at 256 dimensions (+2.4-5.3%); broadly distributed information.
- 2** RoBERTa-InBedder and General-purpose Language Model backbones: larger optimized-random gaps (+8.2-11.8%); stronger information concentration.
- 3** Cross-task transfer correlations are near zero, but transferred selection consistently stays above random (+0.7%-1.6%) across retained dimensions.
- 4** Mechanism: Retrieval-optimized Embedders and most Instruction-conditioned Embedders show higher entropy and flatter top-k concentration curves. RoBERTa-InBedder and General-purpose Language Model backbones show relatively lower entropy and steeper curves.